

Component Derating & DFMEA Desktop Application

1. Overview

This application is a **desktop-based engineering analysis tool** built using the **MERN Stack** + **Electron**, designed to automate:

- **Component Derating Analysis**
- **Design Failure Mode and Effects Analysis (DFMEA)**

The system allows users to upload a **Bill of Materials (BOM)** and perform **automated engineering evaluations** for each component using predefined datasets such as:

- Derating Guidelines
- Potential Failure Modes
- Severity, Occurrence, Detection datasets

It eliminates manual calculations, reduces human error, and ensures **standardized engineering evaluation workflows**.

2. Key Objectives

- Automate **BOM parsing and component-level analysis**
 - Ensure **accurate derating calculations**
 - Enable **structured DFMEA evaluation**
 - Provide **dynamic calculations & validations**
 - Maintain **persistent session-based data storage**
 - Export results into **Excel reports with visual indicators**
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3. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB
Desktop Wrapper	Electron
File Handling	Excel (XLSX parsing & export)

4. System Architecture

High-Level Flow

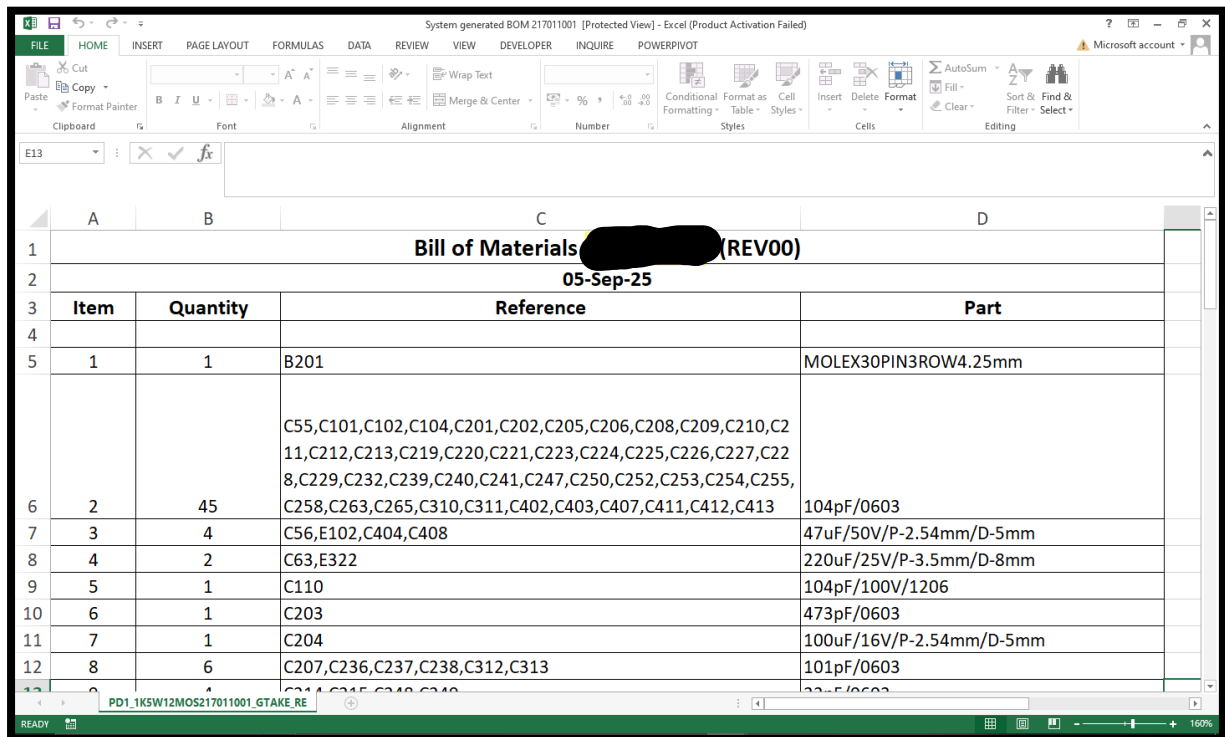
1. User uploads BOM file
2. Backend parses and extracts component data
3. Data stored as a session in MongoDB
4. Frontend dynamically generates pages per component
5. Calculations and validations happen in real-time
6. Final results exported to Excel

5. Core Modules

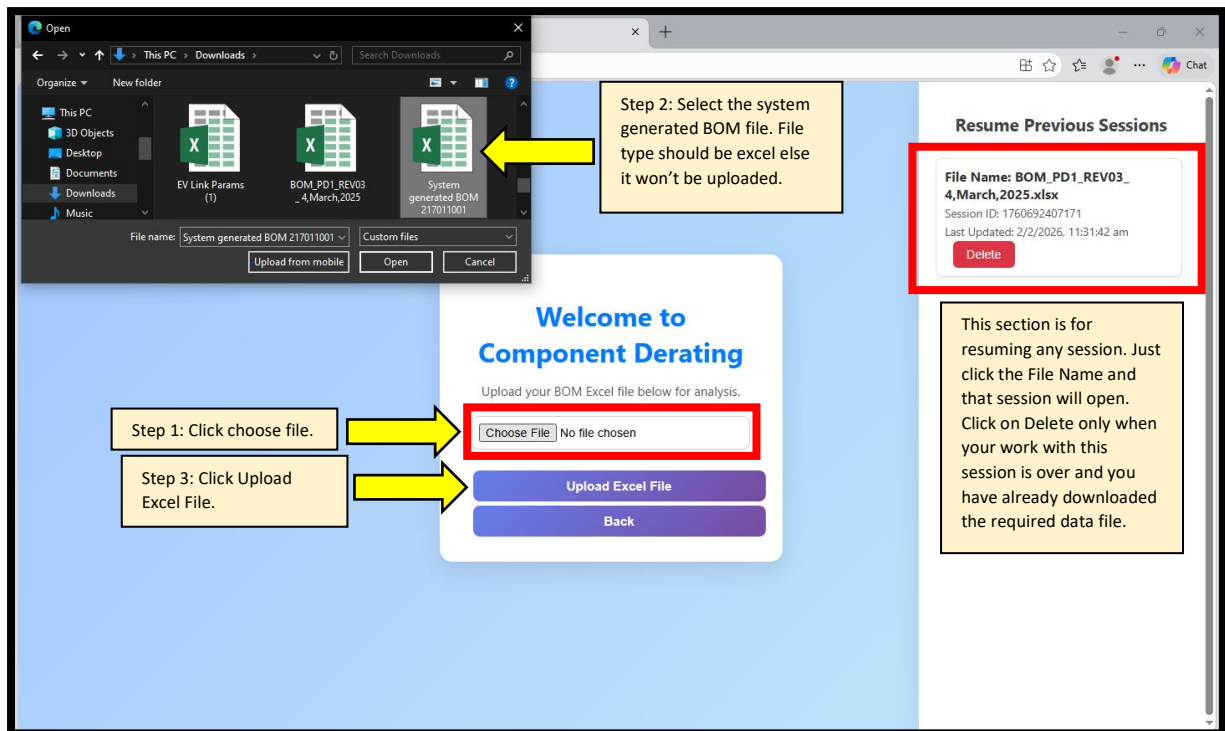
5.1 BOM Upload & Parsing Module

Functionality

- Upload system generated BOM file



Item	Quantity	Reference	Part
1	1	B201	MOLEX30PIN3ROW4.25mm
2	45	C55,C101,C102,C104,C201,C202,C205,C206,C208,C209,C210,C211,C212,C213,C219,C220,C221,C223,C224,C225,C226,C227,C228,C229,C232,C239,C240,C241,C247,C250,C252,C253,C254,C255,C258,C263,C265,C310,C311,C402,C403,C407,C411,C412,C413	104pF/0603
3	4	C56,E102,C404,C408	47uF/50V/P-2.54mm/D-5mm
4	2	C63,E322	220uF/25V/P-3.5mm/D-8mm
5	1	C110	104pF/100V/1206
6	1	C203	473pF/0603
7	1	C204	100uF/16V/P-2.54mm/D-5mm
8	6	C207,C236,C237,C238,C312,C313	101pF/0603



Processing

- Backend parses Excel file
- Maps extracted data into structured JSON
- Stores session data in database

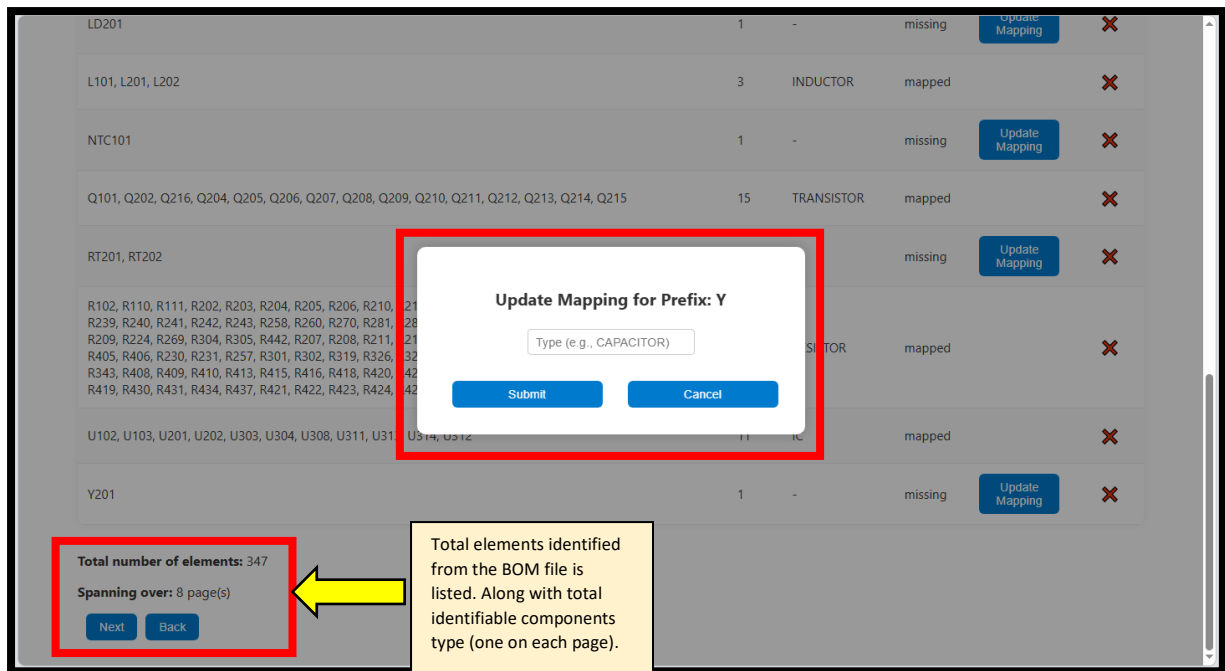
General Instructions

Your file contains the below elements:

Names	Count	Type	Status	Action	Delete
B201	1	-	missing	Update Mapping	×
C55, C104, C201, C202, C205, C206, C208, C209, C210, C211, C212, C213, C219, C220, C221, C223, C224, C225, C226, C227, C228, C229, C232, C239, C240, C241, C247, C250, C252, C253, C254, C255, C258, C263, C265, C237, C414,					×
Derived from the excel file and not matched with the type of component from the database. For that use Update Mapping if you know the type of component. It will be saved in the database for future use.					
E102, E322, E103, E101, E104, E301, E302, E318, E319, E320, E321	11	ELECTROLYTIC CAPACITOR	mapped		×
D101, D106, D107, D107, D202, D213, D303, D304, D306, D307, D308, D309, D305, D402, D403, D404, D405, D406, D407, D408, D409, D414, D410, D411	24	DIODE	mapped		×
F1, F2	2	FUSE	mapped		×
HM102, HM201, HM202, HM302, HM303	7	-	missing	Update Mapping	×
MH1, MH2, MH3	3	-	missing	Update Mapping	×
TP101, TP102, TP201, TP202, TP203, TP204, TP205, TP206, TP207, TP208, TP210, TP211, TP212, TP213, TP214, TP215, TP216, TP217, TP218, TP219, TP220, TP221, TP222, TP223, TP224, TP225, TP226, TP227, TP228, TP229,					

Derived from the excel file and matched with the type of component from the database.

You can click on the cross icon to not include that entire component in your result data file.



5.2 Component Derating Module

Overview

Each BOM component is processed individually with a dedicated UI page.

Fields Captured

1. Type
2. Reference Number
3. Material
4. Rating
5. Derating Parameter
6. Derating Factor
7. Actual Rating (User Input)
8. Actual Value (User Input)
9. Derated Value (Auto-calculated)
10. Remarks (Auto-generated)

Logic & Workflow

CAPACITOR
 Total elements: 84

Component Type and all its elements listed in this page.

Name	Material	Rating
C55	Select Material	104pF/0603
C101	ELECTROLYTIC ALUMINIUM	104pF/0603

Select Material from the dropdown. If the Material of your requirement isn't listed then ask the Admin to add that in the database using the ADMIN portal.

Rating is picked from the BOM itself.

Derating Parameters for ELECTROLYTIC ALUMINIUM

Derating Parameter	Derating Factor	Actual Rating	Actual Value	Derated Value	Remarks
Ambient or Case Temperature	0.9	Actual Rating	Actual Value	-	-
Surge Current	0.8	Actual Rating	Actual Value	-	-
Ripple Voltage or Current	0.8	Actual Rating	Actual Value	-	-
DC Voltage	0.8	Actual Rating	Actual Value	-	-
dv/dt when rated	0.8	Actual Rating	Actual Value	-	-
Temperature from Max Limit	20	Actual Rating	Actual Value	-	-

As soon as you select the material, you'll see a table structure like this. You need to fill in the Actual Rating & Actual Value column. Derated Value and Remarks will automatically be calculated. For any discrepancies in Derating Parameter & Derating Factor values, contact the Admin to make required changes in the database using the ADMIN portal.

C102 Select Material 104pF/0603

Step 1: Auto Mapping

- System matches:
 - Component Type
 - Material
- with **Derating Guidelines** dataset

Step 2: Auto Population

- Derating Parameter
- Derating Factor

Step 3: Dynamic Calculations

Derated Value

Derated Value = Derating Factor × Actual Rating

Remarks Logic

Condition	Output
Actual Value < Derated Value	Suitable
Actual Value > Derated Value	Not Suitable
Missing (*) values	Deferred

Step 4: Input Constraints

- Only numeric values or * allowed
- * indicates unknown value

5.3 DFMEA Module (Design Failure Mode & Effects Analysis)

Overview

Performs structured DFMEA analysis for each component using predefined datasets.

Fields Captured

1. Type
2. Function
3. Failure Modes
4. Failure Effects
5. Severity (Effect, Criteria, Rank)
6. Causes
7. Occurrence (Effect, Criteria, Rank)
8. Prevention
9. Detection
10. Detection Ranking
11. Physical Controls
12. RPN (Auto-calculated)
13. Remarks

Logic & Workflow

Step 1: Failure Mode Mapping

- Based on Component Type
- Fetched from **Potential Failure Modes dataset**

Step 2: Dynamic Ranking System

Each of the following uses **dependent dropdowns**:

Severity

CAPACITOR

Total elements: 84

Name	Material	Rating	Function
C55	CERAMIC	104pf/0603	Function

Severity Occurrence Detection RPN

Potential Failure Modes	Potential Effects of Failure	Effect	Criteria	Rank
Short	Effect of Failure	Select Effect	Select Criteria	
Open	Effect of Failure	Annoyance	Appearance or Audible N	3
20% Tolerance	Effect of Failure	Select Effect	Select Criteria	

C101 Select Material

Severity Occurrence Detection RPN

Potential Failure Modes Potential Effects of Failure

Short Effect of Failure

Open Effect of Failure

20% Tolerance Effect of Failure

Effect of Failure

Select Effect

Select Criteria

Rank

As soon as you select the material, you'll see a table structure like this for Severity, Occurrence & Detection. You need to fill in the text values for Function and other text related fields. For Effect & Criteria, you need to select from the dropdown available. Mind it you need to select Effect's value first and then Criteria's dropdown will be filled with appropriate values as it is dependent on the Effect column. Rank will appear automatically depending on the values chosen in Effect & Criteria column. The options in Effect & Criteria column are hardcoded so for any changes in those options will require a change in software! Apart from that if there exists any new Potential Failure Modes, those can be added by the Admin in the ADMIN panel.

- Effect → Criteria → Rank

Occurrence

- Effect → Criteria → Rank

Detection

- Effect → Criteria → Rank

Step 3: Ranking Calculation

Ranks are auto-fetched based on:

- Selected Effect
- Selected Criteria

Step 4: RPN Calculation

CAPACITOR

Total elements: 84

Name	Material	Rating	Function
C55	CERAMIC	104pF/0603	Function

Severity

Occurrence

Detection

RPN

Potential Failure Modes	RPN	Recommended Actions
Short		Recommended Action
Open	45	Recommended Action
20% Tolerance		Recommended Action

Automatically calculated from the previous three ranks.

C101

Select Material

104pF/0603

Function

Severity

Occurrence

Detection

RPN

Potential Failure Modes	RPN	Recommended Actions
Short		Recommended Action
Open		Recommended Action
20% Tolerance		Recommended Action

C102

Select Material

104pF/0603

Function

RPN = Severity Rank × Occurrence Rank × Detection Rank

Step 5: Risk Evaluation

RPN Value	Indicator
> 300	High Risk (Red)
≤ 300	Acceptable (Green)

5.4 Admin Panel

Welcome Admin

Check Component Mappings

Check Derating Models

Check Material Options

Check Failure Modes

Back

Component Mappings

C

Type: CAPACITOR

EditDelete

D

Type: DIODE

EditDelete

E

Type: ELECTROLYTIC CAPACITOR

EditDelete

F

Type: FUSE

EditDelete

L

Type: INDUCTOR

EditDelete

Q

Type: TRANSISTOR

EditDelete

R

Type: RESISTOR

EditDelete

SW

Type: SWITCH

EditDelete

T

Type: TRANSFORMER

EditDelete

U

Type: IC

EditDelete

Add New Component

Prefix

Type

Add

Check Component Mappings

Check Derating Models

Check Material Options

Check Failure Modes

Back

EditDelete

RESISTOR - WIREWOUND POWER

- Power Dissipation: 0.5
- Temperature from Max Limit: 125

EditDelete

RESISTOR - THICK/THIN FILM

- Power Dissipation: 0.5
- Voltage: 0.8
- Joule Rating for Inrush Limiting Resistors: 0.8
- Temperature from Max Limit: 80

EditDelete

SWITCH - ALL SMT AND THROUGH HOLE

- Resistive Load Current: 0.75
- Operating Cycles: 0.9
- Ambient or Case Temperature: 0.8
- Capacitive Load Current: 0.75
- Inductive Load Current: 0.4
- Contact Power: 0.5

EditDelete

ELECTROLYTIC CAPACITOR - ELECTROLYTIC ALUMINIUM

- Ambient or Case Temperature: 0.9
- DC Voltage: 0.9
- Surge Current: 0.8
- Ripple Voltage or Current: 0.8
- Temperature from Max Limit: 20

EditDelete

DIODE - ULTRAFAST SWITCH

- DC Voltage: 0.7
- Forward Current: 1

EditDelete

Add New Derating Model

TypeMaterial

Parameter

Factor

+ Add ParameterAdd Model

Check Component Mappings

Check Derating Models

Check Material Options

Check Failure Modes

Back

Material Options

CAPACITOR

- CERAMIC
- ELECTROLYTIC ALUMINIUM
- FILM
- TANTALUM

EditDelete

DIODE

- LED
- ZENER
- SIGNAL/SWITCH
- TRANSIENT SUPPRESSOR
- SCHOTTKY/POSITIVE
- POWER RECTIFIER
- ULTRAFAST SWITCH

EditDelete

RESISTOR

- FILM
- THERMISTOR
- COMPOSITION
- WIREWOUND ACCURATE
- WIREWOUND POWER
- THICK/THIN FILM

EditDelete

TRANSISTOR

- SILICON MOSFET
- SILICON BIPOLAR
- GAAS MOSFET

EditDelete

FUSE

- CHIP OR THROUGH HOLE

EditDelete

INDUCTOR

- PULSE TRANSFORMERS
- COILS

EditDelete

SWITCH

- ALL SMT AND THROUGH HOLE

EditDelete

ELECTROLYTIC CAPACITOR

- ELECTROLYTIC ALUMINIUM

EditDelete

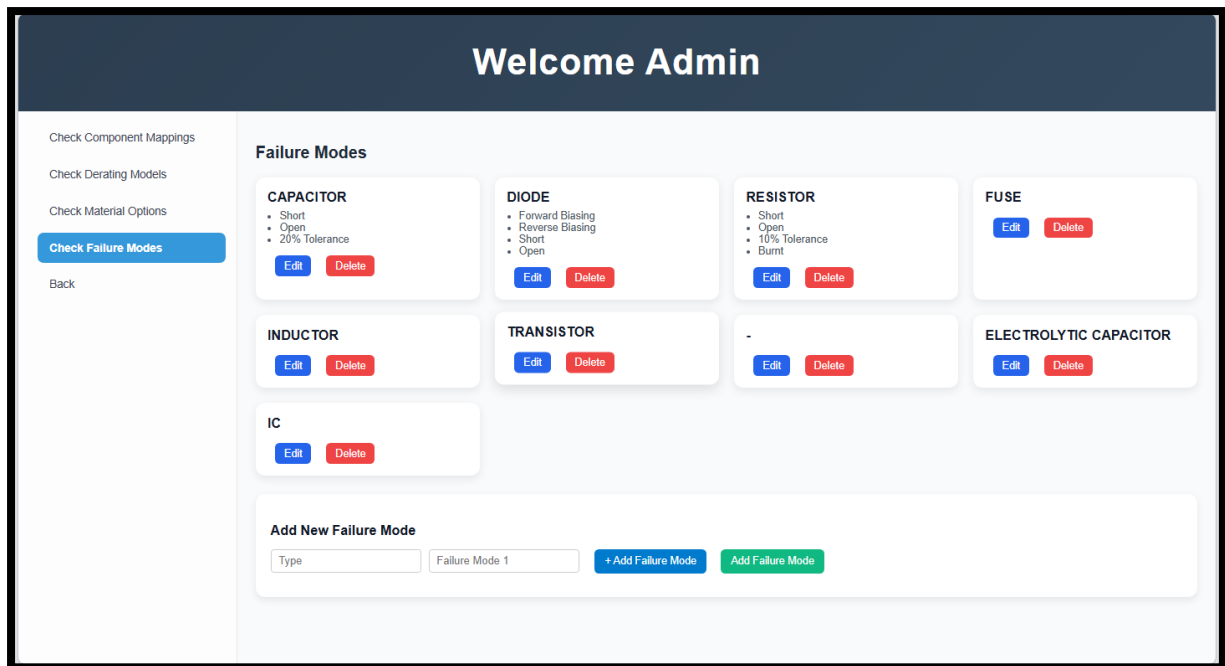
IC

EditDelete

Add New Material Option

TypeMaterial 1

+ Add MaterialAdd Material Option



Purpose

Provides centralized control over datasets used in analysis.

Managed Data

- Derating Guidelines
- Failure Modes
- Severity dataset
- Occurrence dataset
- Detection dataset

Capabilities

- Add / Edit / Delete records
- Maintain consistency of engineering rules
- Enable dynamic dropdown population

5.5 Preview & Export Module

Preview Page

- Displays all component data
- Read-only mode
- Color-coded insights

Export Functionality

- Download full dataset in Excel format
- Maintains:

- Formatting
- Color indicators
- Structured layout

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
1	Type	Component	Material	Function	Failure Mode	Effects	Severity Rank	Causes	Occurrence Rank	Prevention	Detection	Physical Detection	Detection Rank	RPN	Recommended Actions	
2	CAPACITOR	C423	CERAMIC		Short	FUNCTIONAL FAULT	5	-	-	-	-	-	-	-	-	
3	DIODE	D101	-	-	Forward Biasing	Battery Voltage will get shorted	8	Wrong Insertion		1	Visual	Visual	Visual	1	8	-
4																
5																
6																
7																

5.6 Session Management

Features

- Each upload creates a session
- Data stored in MongoDB
- Supports:
 - Resume workflow
 - Back navigation consistency
 - Partial saves

6. Data Flow Summary

BOM Upload → Parsing → DB Storage → Component Pages →
Derating / DFMEA Processing → Preview → Excel Export

7. Key Features

- Automated BOM parsing
- Dynamic UI per component
- Real-time calculations
- Dependent dropdown logic
- Excel export with formatting
- Session persistence
- Admin-controlled datasets
- Desktop deployment via Electron

8. Impact & Benefits

- Reduces manual engineering effort
- Improves accuracy in derating & DFMEA
- Ensures standardized evaluation

- Enhances productivity through automation
- Provides audit-ready reports